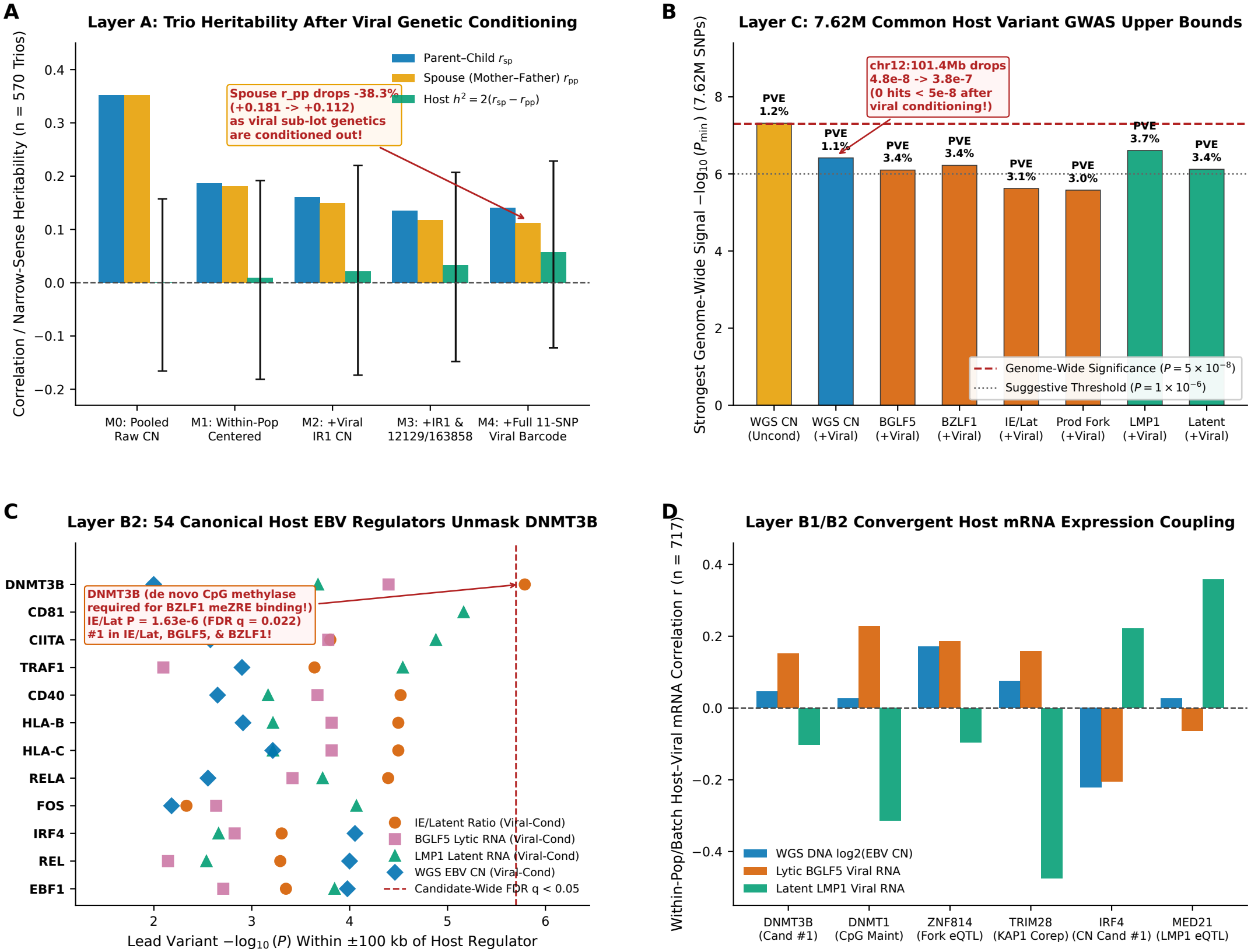
A
Layer A: Trio Heritability After Viral Genetic Conditioning
Correlation / Narrow-Sense Heritability (n = 570 Trios)
Parent–Child r_{sp}
Spouse (Mother–Father) r_{pp}
Host $h^2 = 2(r_{sp} - r_{pp})$
Spouse r_pp drops -38.3%
(+0.181 -> +0.112)
as viral sub-lot genetics
are conditioned out!
M0: Pooled Raw CN
M1: Within-Pop Centered
M2: +Viral IR1 CN
M3: +IR1 & 12129/163858
M4: +Full 11-SNP Viral Barcode
B
Layer C: 7.62M Common Host Variant GWAS Upper Bounds
Strongest Genome-Wide Signal $-\log_{10}(P_{min})$ (7.62M SNPs)
chr12:101.4Mb drops
4.8e-8 -> 3.8e-7
(0 hits < 5e-8 after
viral conditioning!)
PVE 1.2%
PVE 1.1%
PVE 3.4%
PVE 3.4%
PVE 3.1%
PVE 3.0%
PVE 3.7%
PVE 3.4%
Genome-Wide Significance ($P = 5 \times 10^{-8}$)
Suggestive Threshold ($P = 1 \times 10^{-6}$)
WGS CN (Uncond)
WGS CN (+Viral)
BGLF5 (+Viral)
BZLF1 (+Viral)
IE/Lat (+Viral)
Prod Fork (+Viral)
LMP1 (+Viral)
Latent (+Viral)
C
Layer B2: 54 Canonical Host EBV Regulators Unmask DNMT3B
DNMT3B (de novo CpG methylase
required for BZLF1 meZRE binding!)
IE/Lat P = 1.63e-6 (FDR q = 0.022)
#1 in IE/Lat, BGLF5, & BZLF1!
DNMT3B
CD81
CIITA
TRAF1
CD40
HLA-B
HLA-C
RELA
FOS
IRF4
REL
EBF1
IE/Latent Ratio (Viral-Cond)
BGLF5 Lytic RNA (Viral-Cond)
LMP1 Latent RNA (Viral-Cond)
WGS EBV CN (Viral-Cond)
Candidate-Wide FDR q < 0.05
Lead Variant $-\log_{10}(P)$ Within ±100 kb of Host Regulator
D
Layer B1/B2 Convergent Host mRNA Expression Coupling
Within-Pop/Batch Host-Viral mRNA Correlation r (n = 717)
WGS DNA log2(EBV CN)
Lytic BGLF5 Viral RNA
Latent LMP1 Viral RNA
DNMT3B (Cand #1)
DNMT1 (CpG Maint)
ZNF814 (Fork eQTL)
TRIM28 (KAP1 Corep)
IRF4 (CN Cand #1)
MED21 (LMP1 eQTL)